

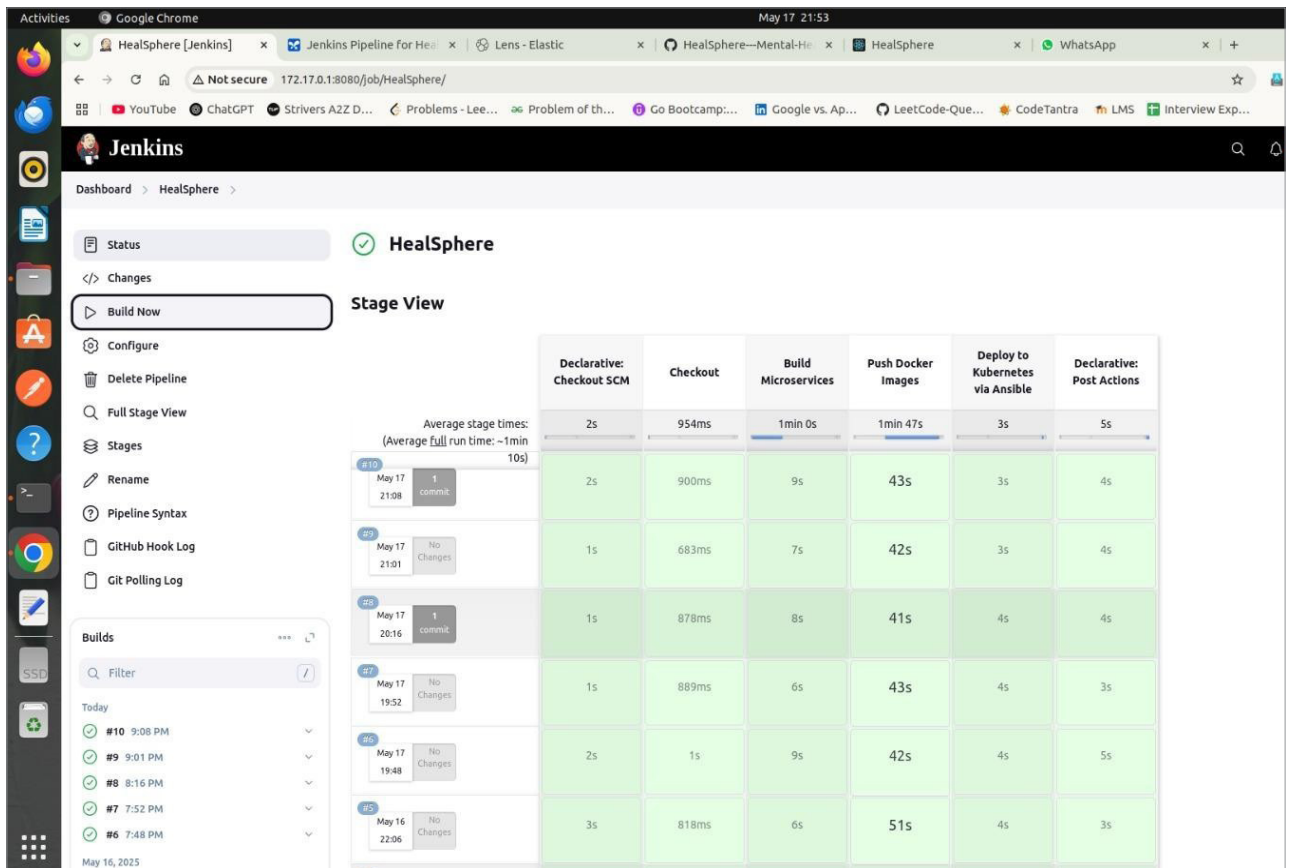


Figure 2: Jenkins Pipeline Stage View for HealSphere CI/CD workflow

User interface highlights

Following deployment, the HealSphere platform becomes accessible via a browser, offering users a clean and role-specific interface. The design focuses on clarity and ease of access to core mental health services.

Dashboard overview: Upon login, users are directed to the main dashboard, which provides centralised access to all modules such as Self-Discovery, Letters, and Crisis Support.

Self-Discovery module: The Self-Discovery module presents users with reflective questions to help them explore their thoughts and emotions. Based on their responses, the platform generates a personalised insights report.

Letters service: The Letters feature provides a safe space for users to write expressive, personal letters. This form of digital journaling supports emotional processing and is saved privately in the user's account.

Crisis Management Support: This module helps users cope with acute emotional stress by offering grounding techniques, calming exercises, and direct access to self-regulation tools.

Self-Help Toolkit: The Self-Help Toolkit offers curated content and interactive exercises focused on themes like self-criticism, emotional regulation, and resilience-building.

Observability with Kibana dashboards

To ensure transparency and real-time insight into system behaviour, HealSphere uses Kibana to visualise logs collected via the ELK Stack. The dashboards help developers and administrators track application events, error trends, and service-specific metrics without having to sift through raw logs manually.

We created modular dashboards tailored to each microservice. These dashboards pull structured data from Elasticsearch, allowing dynamic filtering based on timestamp, namespace, log level, or service name.

This dashboard displays live log data for key services including authentication, appointment management, and the self-help toolkit. Visual filters allow quick identification of failed logins, deployment errors, and user activity spikes.

Key features observed in the dashboard are:

- **Timestamp-based filtering** to view events in specific time windows.
- **Search by log level** (INFO, ERROR, WARN) for debugging purposes.
- **Pod and service metadata** for isolating container-specific issues.
- **Real-time updates** via auto-refresh to monitor deployment events or crash loops.